Pham Hung Quoc Viet

+84 937 404 708 • Thu Duc, Ho Chi Minh City
phamviet1415@gmail.com • github.com/vietpham10205

OBJECTIVE

Proactive 3rd-year student at University of Information Technology – VNU-HCM (UIT) with practical experience in Big Data Processing, Data Warehousing, and Machine Learning. Adept at building robust data pipelines and translating complex datasets into actionable strategic insights. Seeking an internship position as a **Data Analyst, Data Engineer, or Business Analyst** to leverage technical proficiency and drive impactful, data-driven business solutions.

EDUCATION

University of Information Technology – VNU-HCM (UIT) — 2023 – 2027 (expected)
Bachelor of Information Systems — GPA: 8.06/10

SKILLS

Programming	Python, Java, C++, PHP, SQL, JavaScript, HTML5, CSS
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn, Plotly, Statistical Analysis, EDA, Data Cleaning & Wrangling, Excel
Databases	SQL Server, PostgreSQL, MySQL, Oracle
Big Data & BI	Apache Spark (PySpark), Kafka, SSIS, SSAS, Power BI, MDX
AI & Machine Learning	Scikit-learn, TensorFlow, XGBoost, Feature Engineering, Ensemble Learning, Anomaly Detection, SMOTE, SHAP, Data Mining
Tools & Platforms	Docker, Git, GitHub, SAP S/4HANA, Odoo, Kaggle, Streamlit, draw.io, UML, Figma
Others	Agile/Scrum Methodology, IELTS 6.5 (2022)

PROJECTS

NYC Taxi — Real-time Fraud Detection & Fleet Intelligence — 2026
PySpark, Kafka, Docker, PostgreSQL, Streamlit, Pandas, Scikit-learn | GitHub

- **Data Ingestion:** Configured Kafka and Zookeeper clusters to handle trip data. Wrote Python producers to push JSON events at 200 msg/s to a centralized Kafka topic.
- **Stream Processing:** Used PySpark Structured Streaming for micro-batch processing (5,000 offsets/trigger). Created 14 features in real-time using Spark SQL and Pandas.
- **Anomaly Detection:** Built an ensemble model combining Isolation Forest (200 estimators) and MinCovDet to identify abnormal fares. Used strict-voting to minimize false positives.
- **Monitoring & Retraining:** Output results to PostgreSQL and built a Streamlit dashboard for tracking metrics. Implemented model self-retraining every 25 batches.

E-commerce Consumer Behavior Analysis & BI Solution — 2025
SSIS, SSAS, SQL Server, Power BI, MDX, Python, Scikit-learn | GitHub

- **ETL Pipeline:** Developed automated SSIS packages to extract, clean, and migrate raw CSV datasets into a structured SQL Server Data Warehouse.
- **OLAP Modeling:** Designed a Star Schema with Fact tables and 5 dimensions. Built SSAS Multidimensional Cubes and wrote MDX queries for Drill-down and Roll-up analysis.
- **Data Mining:** Implemented K-Means clustering for customer segmentation and Decision Trees to predict spending levels using Python (Scikit-learn).

- **Visualization:** Created interactive Power BI dashboards integrating OLAP cubes and ML predictions to monitor customer retention and purchasing behaviors.

Bank Customer Churn Prediction & Explainability 2025

Python, Scikit-learn, TensorFlow, XGBoost, LightGBM, SMOTE, SHAP, Optuna, Pandas, Seaborn | GitHub

- **EDA & Feature Engineering:** Conducted in-depth EDA on a 10,000-record European bank dataset. Performed data cleaning, Winsorization, one-hot encoding, z-score normalization, and engineered new predictive features to improve model signal.
- **Class Imbalance Handling:** Applied SMOTE to address 80/20 class imbalance (churn rate 20.4%), augmenting training data from 6,400 to 10,192 samples and boosting recall for the minority class.
- **Multi-Model Benchmarking:** Trained and compared 7+ algorithms (Logistic Regression, SVM, Random Forest, XGBoost, LightGBM, CatBoost, AdaBoost) with GridSearchCV and Optuna hyperparameter tuning.
- **Explainability (XAI):** Applied SHAP and LIME to interpret predictions; Jaccard similarity analysis (SHAP vs. LIME: 0.596) identified Age, IsActiveMember, and NumOfProducts as key churn drivers.